‘up yours at the end of life’

Opposition, Emotion, and Overlap in a Corpus of Scottish Debates on Assisted Dying

Marc Alexander and James Balfour · CILC16, Salamanca · May 2025

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1 Content Advisory

- Death (throughout)
- Suicide (throughout)
- Suffering at end of life (throughout)
- Abortion (brief mention in texts)

2 Assisted Dying

- In texts of citizens writing their opinions, for a government audience, on something as controversial as Assisted Dying, what can corpus linguistics bring to the table?
- Three questions:
  - What is linguistically polarized in this debate?
  - How do these polarized ideas lead to contested meanings?
  - What are some societal implications?

3 Outline

- Background
- Our Corpora
- Polarization and Entrenchment
- Argumentation and Authority
- Implications and Democracy
- Conclusion

1. Background



FIGURE 1 BBC News article headlined 'Scottish assisted dying bill passes first vote', with a photo of Liam McArthur MSP holding a banner alongside campaigners with 'Let Us Choose' and 'Assisted Dying' placards outside the Scottish Parliament, 13 May 2025



FIGURE 2 News article headlined 'Assisted Dying debate was Scottish Parliament at its best', with a photo of MSPs including Liam McArthur seated in the Scottish Parliament chamber

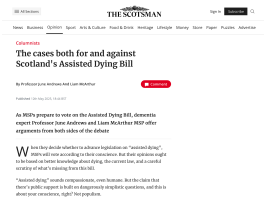


FIGURE 3 The Scotsman opinion column headlined 'The cases both for and against Scotland's Assisted Dying Bill', by Professor June Andrews and Liam McArthur MSP, 12 May 2025



FIGURE 4 The Times comment piece by Sarah Ditum headlined 'Assisted dying can't be sanitised. Stop trying', on MPs' attempts to amend the Terminally Ill Adults (End of Life) Bill



FIGURE 5 The Times column by Iain Macwhirter headlined 'I supported voluntary euthanasia before witnessing my mother's death', published ahead of the Holyrood vote on Liam McArthur's assisted dying bill



FIGURE 6 Protesters, some in white masks, holding placards reading 'Kill the Bill, Not the Ill' and 'Don't Make Doctors Killers', captioned as attempts to change assisted dying law in the Scottish and UK parliaments (Getty Images)



FIGURE 7 Campaigners outside the Scottish Parliament holding 'Let Us Choose' and 'Give Me Choice Over My Death' placards behind a blue banner reading 'Vote Yes to Assisted Dying'



FIGURE 8 Opponents of the bill, including a woman using a wheelchair, holding hand-written placards reading 'Vote Against the Bill', 'Support to live NOT assistance to die', and 'Kill the bill Not me'



FIGURE 9 Dignity in Dying Scotland campaigners in pink t-shirts holding 'Let Us Choose' and 'My Choice, Dignity' placards, with one woman holding a hand-written 'My Choice' sign

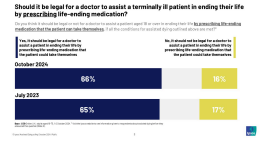


FIGURE 10 CARE website guide page headlined 'Consultation on Assisted Dying For Terminally Ill Adults (Scotland) Bill', quoting Dr Stuart Weir opposing the bill

4 Personal Experience

> For 26 years I worked in the emergency service […] I have witnessed at first hand and close up the sheer desperation and suffering caused by terminal illness. I have had to examine bodies and scenes where the person has died alone and in obvious agony. People with cancer who have vomited and excreted blood and body matter into bowls, coal scuttles, sinks, beds, floors etc.. before finally dying, often over a period of days and weeks […] I was also called to all suicides and lost count of the number of people I have had to cut down from their nooses whilst their grieving family cried in the adjacent room. […] In April 2020 I was diagnosed with inoperable pancreatic cancer and given 6-12 months to live if I chose not to have palliative chemotherapy, and possibly 18 months if I chose to undertake it. My world fell apart and I had to break the news to my wife and two sons. […] I am a proud and private individual who does not want to spend my final days, weeks, or god forbid months, in a hospital or hospice, in pain with no quality of life and having to be lifted off a bed pan or worse! I do not want to be forced into ending my life alone, maybe in my car with the embers of a disposable BBQ smouldering in the boot, or hanging in a lonely wood waiting for some poor unsuspecting member of the public to find me – but with no compassionate alternative that is what I will do.

5 Why Is This Interesting?

- A rich dataset (separate to social media) with citizens writing persuasive texts
- Shows profound disagreements between citizens, and is linguistically highly oppositional
- Broad societal implications of outcomes

6 Current Situation

- September 2021: Assisted Dying for Terminally Ill Adults (Scotland) Bill
- Key differences from previous attempts:
  - Limited to terminally ill, mentally competent adults
  - Scottish residency requirement (12+ months)
  - Requires evaluation by two independent doctors
  - Multiple decision points and waiting periods
  - Self-administration requirement

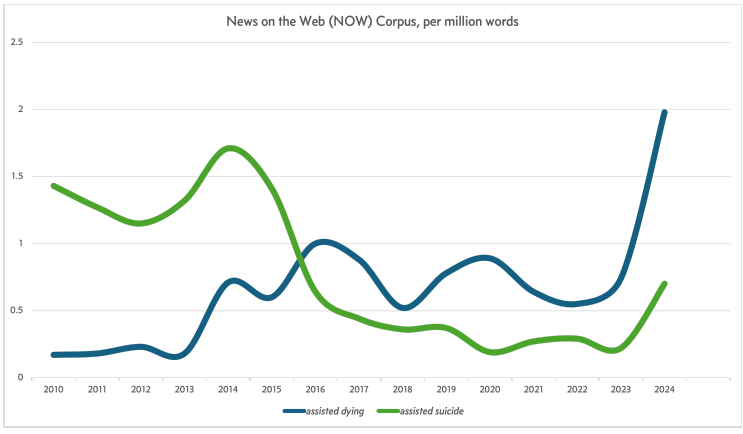


FIGURE 11 Bar chart comparing total consultation word counts by stance (Supportive, Opposed, Neutral/Unsure) for 2021 and 2024, showing supportive submissions dominating in 2021 and opposed submissions dominating in 2024

7 Terminology

No terminology choice is neutral in this debate

- **"Assisted Dying"**: Term used in current bill's title
- **"Assisted Suicide"**: Primarily used by opponents
- **"Euthanasia"**: Not accurate for current bill (euthanasia is physician-administered and this bill requires self-administration)

2. Our Corpora

8 Corpora

- Consultation Corpus
  - Free text entries from two public consultations
  - Metadata includes self-reported closed question on support/opposition to the bill
  - 2021: 2m words across 12,255 written responses
  - 2024: 1.1m words across 7,122 written responses
- Press Corpus
  - Press articles in the UK from the months surrounding the two consultations
  - Obtained through a set of seed search terms
  - 2021: 2.3m words across 1,668 articles
  - 2024: 2.6m words across 2,969 articles

9 Consultations

Two consultation stages:

- **First (2021)**: Conducted by McArthur during bill formulation
  - Largest consultation ever for a Scottish bill (12,255 responses)
- **Second (2024)**: By Health, Social Care and Sport Committee
  - A 'call for views' must take place as part of Stage 1 consideration (7,122 responses)

10 Consultation One

Period: 23 September 2021 – 22 December 2021

Individual available responses: 12,255 Word count: 2,018,421

9,651 fully or partially supportive, 1.2m words 2,554 fully or partially opposed, 0.8m words 50 neutral or unsure, 18k words

Ranges from 0 to 9,759 words per response (average 182, median 99)

11 Consultation Two

Period: 7 June 2024 – 16 August 2024

Individual available responses: 7,122 Word count: 805,755

(1,082,534 when including "Other" responses to multiple-choice questions)

425 fully or partially supportive, 81,353 words 6,681 fully or partially opposed, 714,729 words

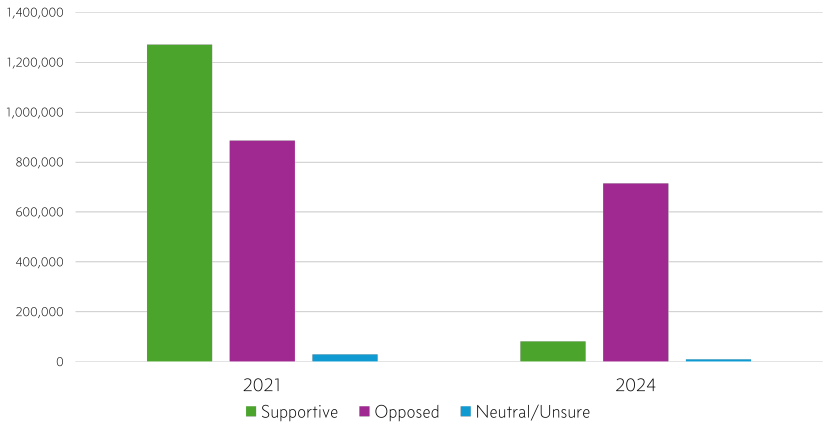


FIGURE 12 Box plot comparing word count per document across six response stance categories (fully opposed, fully supportive, neutral, partially opposed, partially supportive, unsure), showing similar medians but many more high-word-count outliers among fully opposed and fully supportive responses

12 2021 Questions

- Question 1: Support for the proposed Bill
- Question 2: Do you think legislation is required
- Question 3: Views of the proposed process
- Question 4: Views on safeguards
- Question 5: Should a body collect data on assisted dying
- Question 6: Views on conscientious objections
- Question 7: Impact on public finances
- Question 8: Impact on protected characteristics
- Question 9: Impact on sustainability
- Question 10: Additional comments or suggestions

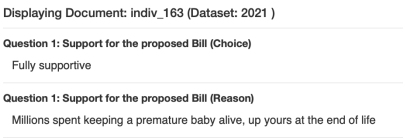


FIGURE 13 Ipsos poll chart showing 66% support and 16% opposition (October 2024) and 65% support and 17% opposition (July 2023) to legalising doctor-assisted dying by prescription

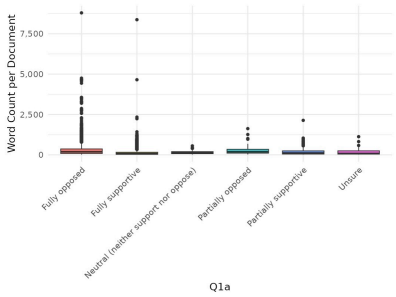


FIGURE 14 Line chart from the NOW Corpus showing frequency per million words of 'assisted dying' and 'assisted suicide' from 2010 to 2024, with 'assisted suicide' peaking around 2014 and 'assisted dying' rising sharply to a high point in 2024

3. Polarization and Entrenchment

13 What is Death?

Collocates of *death*:

2021 Supportive			**2021 Opposed**		
collocation	count	score	collocation	count	score
assisted death	581	3.2	cause death	120	4.38

2021 Supportive			2021 Opposed		
dignified death	198	4.51	natural death	81	4.67
painful death	190	4.1	death (is) final	45	4.55
bad death	59	4.35	early death	42	4.34
manner death	67	3.69	premature death	41	5.3
good death	86	2.64	choose death	51	3.37
undignified death	37	4.06	listed death	31	5.17
peaceful death	39	3.67	physician-assisted death	29	4.86

Supportive	Opposed
Death is something to be **judged** and part of **medical care**?	Death is something with its own **timescale**, a **threshold** to be respected?

14 Key Concepts

- USAS-tagged 2021 corpus (Wmatrix)
- Log-likelihood against BE21 (Paul Baker)

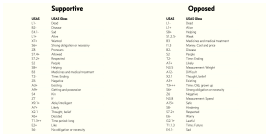


FIGURE 15 Table listing top USAS semantic tag categories and glosses for the Supportive and Opposed subcorpora, e.g. L1- (Dead), B2- (Disease), S8+ (Helping) and S1.2.5- (Weak)

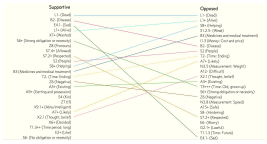


FIGURE 16 Slope chart connecting shared USAS semantic categories between the Supportive and Opposed rankings, showing how far each category's rank shifts between the two subcorpora

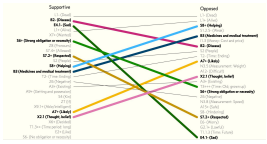


FIGURE 17 The same USAS slope chart with the categories that shift most between the Supportive and Opposed rankings highlighted, including B2- (Disease), S8+ (Helping), A7+ (Likely), X2.1 (Thought, belief), S7.2+ (Respected), and E4.1- (Sad)

15 Q1 USAS ('key concepts') against BE21

- **Supportive** top concepts:
  - L1- (Dead), B2- (Disease), E4.1- (Sad), L1+ (Alive), X7+ (Wanted)
  - S6+ (Strong obligation or necessity), Z8 (Pronouns), S7.4+ (Allowed)
  - S7.2+ (Respected), S2 (People), S8+ (Helping)
- **Opposed** top concepts:
  - L1- (Dead), L1+ (Alive), S8+ (Helping), S1.2.5- (Weak)
  - B3 (Medicines and medical treatment), I1.3 (Money: Cost and price)
  - B2- (Disease), S2 (People), T2- (Time: Ending), A7+ (Likely)

16 2021 "Easy Read Consultation Summary"

"Over three quarters supportive"

"Lots of these people thought:

- it would be kind to let adults who are terminally ill have a safe assisted death if they wanted it.
- it would stop their pain and suffering.
- assisted dying should be allowed in Scotland because it's allowed in some other countries."

17 Personal Experience: Supporting

"It's about choice. Currently, other people's choices prevent those who are terminally ill from dying when they wish to. That's cruel and unfair."

"It's about compassion and autonomy. I have medical knowledge and know that some natural deaths can never be made 'good'. Drugs have their limits."

"We will surely look back and wonder why it took so very long and why so many people had to yield to a process which denies them control, autonomy and dignity, because we were not brave enough to approach this subject in a calm, rational and compassionate way."

"I had to watch my dad die of pancreatic cancer."

"I was fully prepared to put myself at risk and buy him drugs from the dark web so that he could stop the pain. However, my daddy told me it was too risky for me, as I work in the legal sector. So I didn't take it any further. But in what world should this be allowed to happen?"

"I'm sitting here in tears thinking about him. I just wish I hadn't listened to him and got him the drugs. Even in the hospice he was still saying if there was an injection or a pill, he would take it. Nobody should need to go through this. It's killing me. He should have been able to leave on his own terms. He shouldn't have had to suffer and waste away. We don't treat dogs like this."

18 2021 "Easy Read Consultation Summary"

"Just under a quarter not supportive"

"Some people thought:

- it was wrong for a person to take their own life, or to get help to end their life.
- it wouldn't be safe because you wouldn't know for sure if the person was making the choice for themselves.
- older people or disabled people might feel worried about other people trying to choose for them. They might feel less important."

19 Religious Perspective: Opposing

"This is a radical change attacking the very roots of welfare and wellbeing – and trust – and would change society."

"We were created in the Image of God – and if this goes forward, we are challenging LIFE – and VALUES and STANDARDS. Does LIFE mean we have a new meaning of LIFE? Parts of society decided to take life in the womb before life was born into the world – now some want to end life before life is over."

"One day each of us has to stand before Almighty God – and answer for the decisions we have made – whether we believe it or not – whether we want to or not."

20 "Slippery Slope" Concerns

"Thus Belgium's initial law in 2002 expanded from almost exclusively terminally ill cancer patients to patients with psychological illnesses such as severe depression or personality disorder and more recently to competent minors. In Canada, the 2016 law which allowed medical assisted dying for physical health reasons was expanded in 2021 to include patients with mental health conditions."

"In Oregon, 5% of patients dying by assisted suicide gave financial reasons for their decision. Those giving as the reason the 'becoming a burden on family, friends or caregivers' increased from 13% in 1998 to 55% in 2017."

"The British Geriatrics Society, the Association for Palliative Medicine of Great Britain and Ireland, the majority of the British Medical Association members who are involved in palliative care or geriatric medicine oppose the legalisation of assisted dying."

21 Unique lexis in supportive subcorpus

suffering

– abject, agonies, brutal, crying, hideous, starve, tears, thirst, torment, torture, traumatising, tumours

humiliation

– bladder, cleaned, drip, eating, humiliation, incontinence, limbs, nappy, urine, vegetable, wash, wearing

prolonging

– drag, dragged, endless, endlessly, lasted, linger

"I have seen first hand the unnecessary pain and torment that is caused by this horrific disease."

"If you'd seen your Parent with advanced Alzheimer's, needing to be fed, washed, nappy changed, YOU WOULDN'T NEED TO ASK!"

22 "watch* my"

```
positive for Huntington's Disease I've |  watched my   | 3 older siblings deteriorate slowly
                                     I |  watched my   | dad die slowly and painfully
                                 I sadly |  watched my   | young brother suffer horrendous pain
            position More than you know |  Watching my  | lovely strong daddy die broke
                                     I |  watched my   | father die a very horrible
                                     I |  watched my   | mother die horribly from terminal
                                     I |  watched my   | daughter die from cancer with
                       When I was 19 I |  watched my   | mother waste away and die
                                     I |   watch my    | father die of cancer slowly
                                I had to |   watch my    | mother suffer excruciating daily pain
                                     I |  watched my   | mum suffer a prolonged and
          before the experience we had |  watching my  | dad suffer for months
                                     I |  watched my   | husband in agony for months
```

23 "first hand"

```
pain endured I have whatched | first hand | my granny suffer greatly and
         adult life I have witnessed | first hand | many deaths with many causes
               and MND and have seen | first hand | their suffering at not being
              should end I have seen | first hand | my Father endure an unnecessarily
                living is I have had | first hand | experience of the long protracted
           living I have witnessed at | first hand | the pain and suffering experienced
           family not machines I have | first hand | experience of deaths that were
   consultation document I have seen | first hand | a relative age 55 who
                  so I have seen at | first hand | the physical and psychological damage
```

24 Unique lexis in opposed subcorpus

bill as slippery slope

- eroded, implies, message, normalisation, normalise, opens, promoting, reinforce, sends, signal, subtle, wedge, widen, widened

value of human life

- burden, burdensome, contradiction, devalue, undermine, undermines

appealing to authority

- American, BMJ, cited, commandments, God's, journal, Mr, professor, shalt, thou

> "The Bill would send a signal that some lives are no longer worth living."
>
> "We are also disobeying one of the 10 commandments which states 'thou shalt not kill'."

25 "clear message"

```
            society This bill sends a | clear message | that some lives have actually
                  and it would send a | clear message | to our frail elderly and
             Assisted suicide sends a | clear message | to frail elderly disabled people
   suffering Assisted suicide sends a | clear message | to frail elderly and disabled
            suffering It also sends a | clear message | to frail elderly and disabled
             Assisted suicide sends a | clear message | to frail elderly and disabled
   suffering Assisted suicide sends a | clear message | to frail elderly and disabled
   suffering Assisted suicide sends a | clear message | to frail elderly and disabled
             Assisted suicide sends a | clear message | to frail elderly and disabled
             Assisted suicide sends a | clear message | to frail elderly and disabled
```

26 Comparative USAS, Log Ratio

Supportive vs Opposed:

- L2 (Living creatures: animals) - Dogs are 'treated better'
- N3.8- (Speed: Slow) - Prolonged death
- B4 (Cleaning and personal care) - Personal narratives of humiliation

Opposed vs Supportive:

- N3.8 (Measurement: Speed) - Slippery slope re: rates
- N3.5 (Measurement: Weight) - Pressure on elderly/sick

- I1.3 (Money: Cost and price) - Danger of 'saving money'
- Z3 (Other proper names) - Appeal to authority

27 Overall

Supportive	Opposed
animals	slippery slope
prolonged death	danger of 'saving money'
personal narratives of suffering and humiliation	appeals to authority
logical argumentation	too soon
experiential-based	**principle-based**

28 5. Implications and Democracy

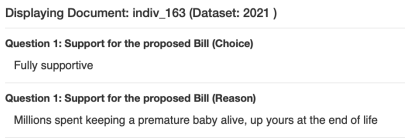


FIGURE 18 Press Coverage Chart

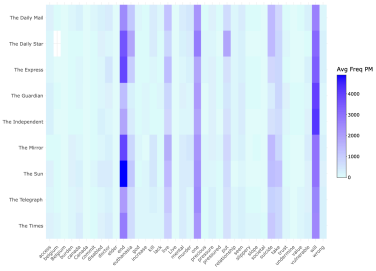


FIGURE 19 Press Coverage Comparison

29 Mail on Sunday (26 December 2023)

> CRISPIN COLEGATE, Pontardawe, Neath Port Talbot. I watched my mother-in-law slowly dying for nearly a year, kept alive by medicines and morphine. She asked me to help her die and I nearly did. Looking back, I wish I had helped her, it would have been worth a prison sentence.
>
> KEITH McCREADY, Bishop's Stortford, Herts. IF I had let any of my pets suffer as Sarah Vine's grandmother did in her last days (Mail), I would be prosecuted for cruelty.

30 Mirror (23 October 2021)

> Critics fear it could leave vulnerable people exposed to unwanted pressure. Opponents include the Archbishop of Canterbury, the Most Rev Justin Welby. He said that while the safeguards in the bill are stronger than in previous attempts, they still do not go far enough. "What we want is assisted living, not assisted dying," he said.

31 Political Opinions

Traditionally:

"Left Wing"	"Right Wing"
Collective protection of the weak and vulnerable in society	Individual responsibility and personal freedom
Progressive values	Conservative values

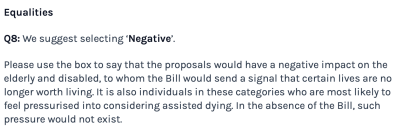
But:

- Discourse **opposing** AD is about collective protection of the weak
- Discourse **supporting** AD is about personal responsibility and freedom

32 Most Frequent Shared Sentences

Shared Sentence Text	Docs	Supportive	Opposed	Neutral
no safeguards are good enough.	26	0	26	0
creating a body responsible for reporting and collecting data can not address possible abuse.	25	0	24	1
terminal illness diagnoses can be unreliable.	24	0	23	1
this is coercive and has no place in a civilised society.	23	0	22	1
in the absence of the bill, such pressure would not exist.	22	0	22	0
life is precious.	21	0	21	0
life is sacred.	21	0	21	0

33 Copy-Paste Evidence


Equalities

Q8: We suggest selecting '**Negative**'.

Please use the box to say that the proposals would have a negative impact on the elderly and disabled, to whom the Bill would send a signal that certain lives are no longer worth living. It is also individuals in these categories who are most likely to feel pressurised into considering assisted dying. In the absence of the Bill, such pressure would not exist.

FIGURE 20 Suggested answer wording



FIGURE 21 Source campaign guide

Two responses (3,758 and 3,991 words) with near-identical text

34 Word Count Comparison 2021 vs 2024

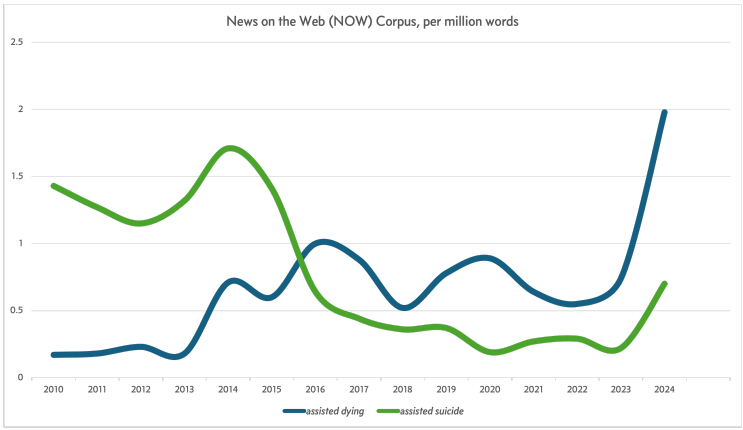


FIGURE 22 Word count comparison

35 Our Questions

We began with three questions:

1. What is linguistically polarized in this debate?

2. How do these polarized ideas lead to contested meanings?
3. What are some societal implications?

36 Elusive Consensus

These linguistic polarisations show societal and ethical divisions. The use of a fundamentally different ethical language means three things are contested in these debates:

- **Contested conceptualization of death**: is death conceptualized as part of medical care (to be optimised) or as a moral threshold (to be respected)?
- **Contested linguistic authority**: experiential-based versus principle-based language: "I watched my mother die" versus "life is always sacred"/"doctors should never kill"; choice/dignity/decide versus vulnerable/pressure/safeguards; crafting individual moral reasoning versus amplifying that of others.
- **Contested meaning of the vulnerable**: those facing painful death or those facing pressure to end their own lives?

37 "up yours at the end of life"

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May 2025

38 Topic Modelling

- Structural Topic Model (STM) fitted to the combined free-text responses from the 2021 and 2024 consultations
- Respondents classified as 'Supportive' or 'Opposed' based on their answers to initial multiple-choice question
- Preprocessing (lowercasing, removing punctuation/numbers/stopwords, stemming)
- Modelled using STM to identify 60 distinct topics

39 Topic Model Results

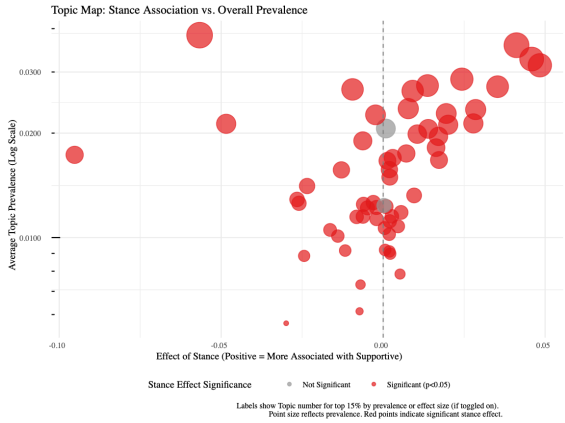
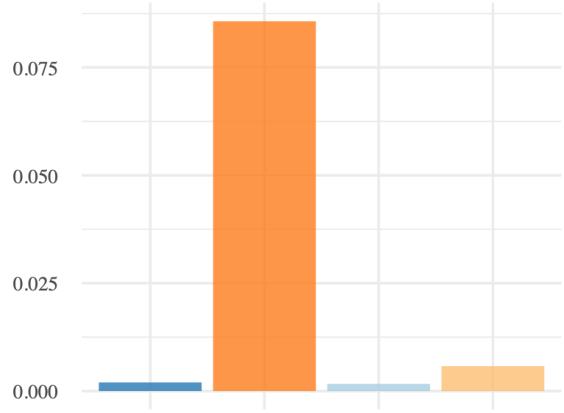


FIGURE 23 Scatter plot titled 'Topic Map: Stance Association vs Overall Prevalence', plotting all 60 STM topics by effect of stance and average prevalence, with topics significantly associated with a stance shown in red

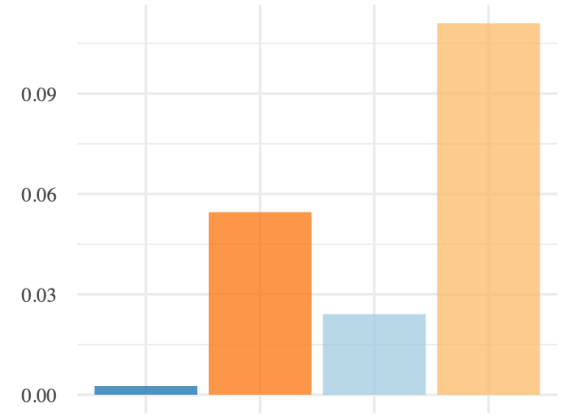
40 Topic 8



8: live, elder, worth, pressu, end**

FIGURE 24 Bar chart of Topic 8's ('live, elder, worth, pressure, end') mean prevalence across four consultation subcorpus segments (2021/2024 supportive/opposed), highest among opposed responses

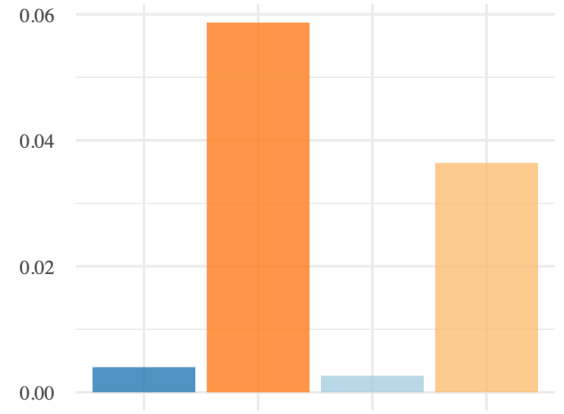
41 Topic 32



32: kill, god, take, murder, one, wrong, precious

FIGURE 25 Bar chart of Topic 32's ('kill, god, take, murder, wrong, precious') mean prevalence across four consultation subcorpus segments (2021/2024 supportive/opposed), highest among opposed responses and most prominent in 2024

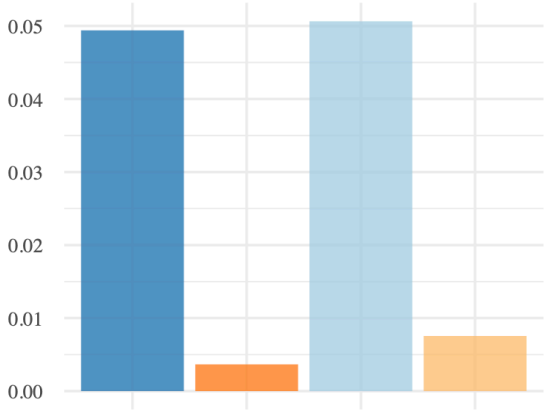
42 Topic 50



50: vulner, pressure, burden, elder, disabl***

FIGURE 26 Bar chart of Topic 50's ('vulnerable, pressure, burden, elderly, disabled') mean prevalence across four consultation subcorpus segments (2021/2024 supportive/opposed), highest among opposed responses

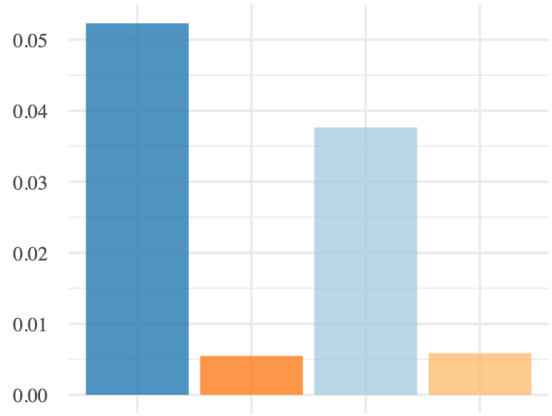
43 Topic 29



29: watch, die, cancer, mother, pass, let

FIGURE 27 Bar chart of Topic 29's ('watch, die, cancer, mother, pass, let') mean prevalence across four consultation subcorpus segments (2021/2024 supportive/opposed), highest among supportive responses in both years

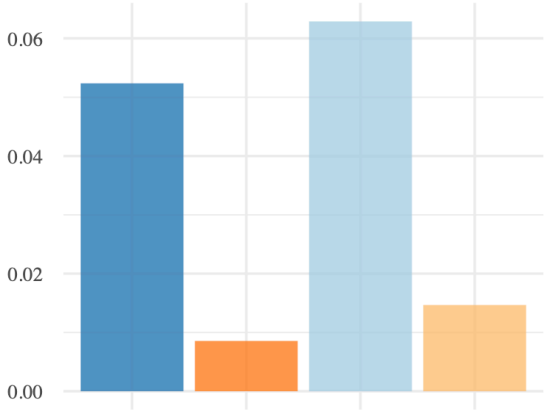
44 Topic 40



40: choice, right, choose, dignity, person, dignif*

FIGURE 28 Bar chart of Topic 40's ('choice, right, choose, dignity, person, dignified') mean prevalence across four consultation subcorpus segments (2021/2024 supportive/opposed), highest among supportive responses

45 Topic 16



16: suffer, pain, one, love, peace, hope, distress

FIGURE 29 Bar chart of Topic 16's ('suffer, pain, one, love, peace, hope, distress') mean prevalence across four consultation subcorpus segments (2021/2024 supportive/opposed), highest among supportive responses

46 Press Corpora

UK press articles which meet the following criteria:

- Available on LexisNexis.
- Between 23 March and 22 December 2021 or 7 December 2023 and 16 August 2024.
- Matching search terms "assisted suicide" OR "assisted dying" OR "euthanasia" OR "assisted to die" OR "assisted death" OR "assisted in dying" OR "assisted end to life" OR "death with dignity" OR "die with dignity" OR "die in dignity" OR "right to die" OR "end of life choice" OR "end of life assistance"

47 Terminology: "Assisted Dying"

- Term used in current bill's title
- Prevalent in consultations and media
- Primarily used by supporters
- Distinguished from suicide by supporters of the term as "people in question are not choosing death over a healthy life"
- Relatively new term (last 15 years in UK)
- Can emphasise disease/death "process" underway

48 Terminology: "Assisted Suicide"

- Primarily used by opponents
- Argued to be more legally/practically accurate
- Critics say "assisted dying" is a euphemism
- Retains the moral/religious stigma of "suicide"
- Can emphasise individual's action

49 Terminology: "Euthanasia"

- Greek "good death"
- Not technically accurate for current bill as euthanasia is physician-administered and this bill requires self-administration

50 NOW Corpus: "Assisted Dying" vs "Assisted Suicide"

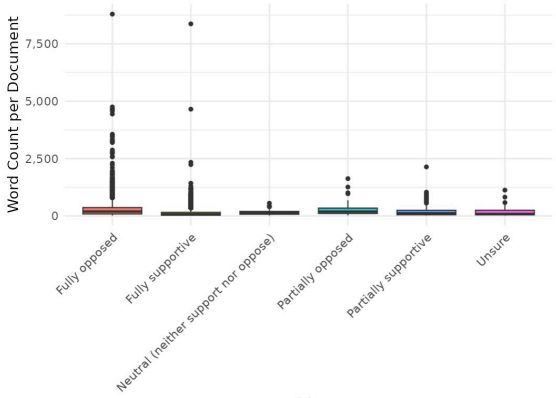


FIGURE 30 NOW Corpus comparison